Charani

Log IQ

Intelligent Log Analysis System

Final Project Presentation

CS 6030-120: Natural Language Processing · Western Michigan University · Spring 2026

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The Problem

- ⚠ 99% of logs remain unexamined
- ⚠ 70-80% SRE time in reactive firefighting
- ⚠ No semantic understanding of incidents

100K+	<1s	94.3%	99.99%
Logs/min	Latency	NLP Acc.	Uptime

Course Tasks · 100 pts

01	02	03
POS Tagging	N-gram LM	NLP Ethics

04	05
Progress Pres	Final Pres.

20 pts each

Three-Layer NLP Pipeline

Charani

Task 01: POS Tagging · Task 02: N-gram LM · Full architecture

Layer 1
Text Understanding

- Tokenization & Preprocessing
- POS Tagging — spaCy v3.7
- NER — 8 Entity Types
- 384-dim Semantic Embeddings
- BERT Severity Classification
- N-gram Baseline (Trigram)

```
POS Sample: "ERROR" → NOUN/SEVERITY · "nginx-7f8" → NOUN/POD_NAME
```

F1: 0.92 · 50ms/log

Layer 2
Language Intelligence

- Fine-tuned GPT (LoRA/LLaMA-2)
- Chain-of-Thought Prompting
- Template-based NL Generation
- Multi-document Summarization
- Semantic Anomaly Detection
- Isolation Forest (2.5σ)

Loss: 0.23 · Acc: 94.3%

Layer 3
Conversational Interface

- RAG — Qdrant Vector DB
- LangChain Orchestration
- Multi-turn Dialogue (LlamaIndex)
- WebSocket Real-time (Socket.IO)
- Grounding & Citation
- Hallucination Guard (<0.75)

Recall: 0.96 · <200ms

Task 03 — NLP Ethics

Rakesh

Responsible AI practices embedded throughout LogIQ

Privacy & PII

NER-based PII redaction in Loki before storage. AES-256 at rest, TLS 1.3 in transit. CloudTrail immutable audit logs.

Model Bias

BERT/GPT trained on general text may misclassify domain-specific patterns. Mitigated via LoRA fine-tuning on 5K+ incident reports.

Hallucination Risk

Confidence threshold <0.75 triggers fallback. Every answer grounded with citations from the RAG vector store.

Transparency & Explainability

Chain-of-thought prompting makes reasoning visible. Every incident diagnosis cites source logs. Auditable via CloudTrail.

Access & Fairness

Natural language interface removes CLI barrier — non-technical stakeholders can query logs in plain English.

Over-reliance Risk

Human-in-the-loop escalation enforced for P1 incidents via PagerDuty. AI suggestions never auto-remediate production.

Cloud NLP — AWS EKS Deployment

Rakesh

Multi-region · Zero Trust security · GitOps CI/CD · Full observability

us-east-1	eu-west-1	ap-southeast-1
PRIMARY	**SECONDARY**	**TERTIARY**
Active / Read-Write	Passive / Standby	Read-Only

7-Layer Architecture

1. **User Interface**
   React · WebSocket · Material-UI · CloudFront/S3
2. **API Gateway**
   Amazon API Gateway · Cognito · WAF · Rate Limiting
3. **Application Services**
   ECS Fargate · FastAPI Microservices
4. **NLP Pipeline**
   SageMaker · BERT · GPT · RAG · Three-Layer
5. **Data Persistence**
   RDS · ElastiCache · OpenSearch · Qdrant
6. **Queue & Streaming**
   Amazon MSK (Kafka) · Kinesis · Exactly-once
7. **Observability**
   CloudWatch · X-Ray · Grafana · Prometheus · Jaeger

RTO < 15 min · RPO < 15 min · Route 53 GeoDNS · Promotion time ~60-90s · WAL streaming replication

Performance & Research Contributions

Pujitha

Benchmarks · Innovations · Course task alignment

100K+	<1s	94.3%	0.92	10K	99.99%
Logs/min	P99 Latency	NLP Accuracy	NER F1 Score	Concurrent Users	Availability

Semantic Log Clustering

BERTopic vs regex on 10M+ Kubernetes events — 34% better cluster purity

RAG for Root Cause Analysis

Hybrid retrieval + citation accuracy. Hallucination guard: confidence threshold <0.75

Anomaly Detection in Embeddings

Isolation Forest on 384-dim embeddings · 2.5σ threshold · 7-day baseline

Fine-Tuned Domain Models

LoRA on 5K+ incident reports · GPT-3.5 vs LLaMA-2 · Parameter efficiency

Course alignment:

POS Tagging 20 pts	N-gram LM 20 pts	NLP Ethics 20 pts	Progress Pres 20 pts	Final Pres. 20 pts

Pujitha

Log IQ

DEMO

Questions?

From Research to Production

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Live Demo Checklist

- Log ingestion — Kafka stream live
- POS tagging on real Kubernetes log
- NER entity extraction (8 types)
- BERT severity classification
- N-gram anomaly score output
- RAG chatbot: natural language query
- dashboard — live metrics
- PII redaction demo (Ethics task)